

Unified Notes on Graph Reductions

Definitions, Examples, and Exam-Ready Iff Proofs

How to Read These Notes

Goal

These notes are arranged exactly in two parts.

- a. Define the decision problems and give small examples.
- b. Prove the important pairs using polynomial reductions and both directions of the iff proof.

Main Reductions

$\text{VERTEX-COVER} \leq_p \text{INDEPENDENT-SET}$, $\text{INDEPENDENT-SET} \leq_p \text{VERTEX-COVER}$, $\text{INDEPENDENT-SET} \leq_p \text{3-SAT}$

$\text{VERTEX-COVER} \leq_p \text{SET-COVER}$, $\text{3-SAT} \leq_p \text{INDEPENDENT-SET}$.

The phrase “equally hard” means the problems are connected by polynomial reductions. In each direct reduction below, the key statement is always

old instance is yes $\iff$ new instance is yes.

Exam Writing Rule

For each pair, write in this order:

1. state the source decision instance;
2. construct the target decision instance;
3. prove the forward direction;
4. prove the backward direction;
5. say the construction is polynomial time.

Do not only describe the idea. Always write the iff statement.

1 Part A: Decision Problem Definitions

1.1 Vertex Cover

Decision Version: VERTEX-COVER

Input:

$$\langle G, k \rangle,$$

where $G = (V, E)$ is an undirected graph and k is an integer.Question: Does G contain a set $C \subseteq V$ such that

$$|C| \leq k$$

and every edge of G has at least one endpoint in C ?Such a set C is called a **vertex cover**.**Example**Let G be the path

$$a - b - c.$$

The set

$$C = \{b\}$$

is a vertex cover, because the edges (a, b) and (b, c) both touch b . Therefore

$$\langle G, 1 \rangle \in \text{VERTEX-COVER}.$$

But if G is a triangle K_3 , then no one vertex covers all three edges, so

$$\langle K_3, 1 \rangle \notin \text{VERTEX-COVER}.$$

1.2 Independent Set**Decision Version: INDEPENDENT-SET**

Input:

$$\langle G, k \rangle,$$

where $G = (V, E)$ is an undirected graph and k is an integer.Question: Does G contain a set $S \subseteq V$ such that

$$|S| \geq k$$

and no two vertices in S are adjacent?Such a set S is called an **independent set**.**Example**Again let G be the path

$$a - b - c.$$

The set

$$S = \{a, c\}$$

is independent, because there is no edge directly between a and c . Therefore

$$\langle G, 2 \rangle \in \text{INDEPENDENT-SET}.$$

But a triangle K_3 has no independent set of size 2, since every pair of vertices is adjacent.

1.3 Clique

Decision Version: CLIQUE

Input:

$$\langle G, k \rangle,$$

where $G = (V, E)$ is an undirected graph and k is an integer.

Question: Does G contain a set $Q \subseteq V$ such that

$$|Q| \geq k$$

and every two distinct vertices in Q are adjacent?

Such a set Q is called a **clique**.

Example

In a triangle K_3 , all three vertices are pairwise adjacent, so

$$\langle K_3, 3 \rangle \in \text{CLIQUE}.$$

For the path $a - b - c$, there is no clique of size 3, because a and c are not adjacent.

1.4 Set Cover

Decision Version: SET-COVER

Input:

$$\langle U, \mathcal{S}, k \rangle,$$

where U is a finite universe, $\mathcal{S}$ is a family of subsets of U , and k is an integer.

Question: Are there at most k sets from $\mathcal{S}$ whose union equals U ?

Example

Let

$$U = \{1, 2, 3, 4\}$$

and

$$\mathcal{S} = \{\{1, 2\}, \{2, 3\}, \{3, 4\}, \{4\}\}.$$

With $k = 2$, the sets

$$\{1, 2\} \quad \text{and} \quad \{3, 4\}$$

cover all elements of U . Hence this is a yes-instance of SET-COVER.

1.5 3-SAT

Decision Version: 3-SAT

Input: a Boolean formula

$$\phi = C_1 \wedge C_2 \wedge \cdots \wedge C_m$$

in 3-CNF, where each clause C_j is an OR of exactly three literals.

Question: Is there a truth assignment to the variables that makes every clause true?

Example

The formula

$$\phi = (x_1 \vee \neg x_2 \vee x_3) \wedge (\neg x_1 \vee x_2 \vee x_3)$$

is satisfiable. For example, set

$$x_1 = 1, \quad x_2 = 1, \quad x_3 = 0.$$

Then the first clause is true because $x_1 = 1$, and the second clause is true because $x_2 = 1$.

2 Part B: Pairwise Reductions and Iff Proofs

3 Vertex Cover and Independent Set

Key Lemma

Let $G = (V, E)$ be an undirected graph and let $C \subseteq V$. Then

$$C \text{ is a vertex cover} \iff V - C \text{ is an independent set.}$$

Proof of the Lemma

Suppose C is a vertex cover. If two vertices $u, v \in V - C$ were adjacent, then the edge (u, v) would have neither endpoint in C . That contradicts the fact that C covers every edge. Hence $V - C$ is independent.

Conversely, suppose $V - C$ is independent. Then no edge has both endpoints in $V - C$. Therefore every edge has at least one endpoint in C , so C is a vertex cover.

3.1 VERTEX-COVER Reduces to INDEPENDENT-SET

Reduction

Take an instance

$$\langle G, k \rangle$$

of VERTEX-COVER. Let $n = |V(G)|$. Output the INDEPENDENT-SET instance

$$\langle G, n - k \rangle.$$

This construction is clearly polynomial time.

Iff Correctness

We prove

$$\langle G, k \rangle \in \text{VERTEX-COVER} \iff \langle G, n - k \rangle \in \text{INDEPENDENT-SET}.$$

First suppose $\langle G, k \rangle \in \text{VERTEX-COVER}$. Then G has a vertex cover C with $|C| \leq k$. By the lemma, $V - C$ is an independent set. Also,

$$|V - C| = n - |C| \geq n - k.$$

Thus $\langle G, n - k \rangle \in \text{INDEPENDENT-SET}$.

Conversely, suppose $\langle G, n - k \rangle \in \text{INDEPENDENT-SET}$. Then G has an independent set S with $|S| \geq n - k$. By the lemma, $V - S$ is a vertex cover. Also,

$$|V - S| = n - |S| \leq k.$$

Thus $\langle G, k \rangle \in \text{VERTEX-COVER}$.

3.2 INDEPENDENT-SET Reduces to VERTEX-COVER

Reduction

Take an instance

$$\langle G, k \rangle$$

of INDEPENDENT-SET. Let $n = |V(G)|$. Output the VERTEX-COVER instance

$$\langle G, n - k \rangle.$$

Iff Correctness

We prove

$$\langle G, k \rangle \in \text{INDEPENDENT-SET} \iff \langle G, n - k \rangle \in \text{VERTEX-COVER}.$$

If G has an independent set S with $|S| \geq k$, then $V - S$ is a vertex cover by the lemma, and

$$|V - S| = n - |S| \leq n - k.$$

Therefore $\langle G, n - k \rangle \in \text{VERTEX-COVER}$.

Conversely, if G has a vertex cover C with $|C| \leq n - k$, then $V - C$ is an independent set by the lemma, and

$$|V - C| = n - |C| \geq k.$$

Therefore $\langle G, k \rangle \in \text{INDEPENDENT-SET}$.

Example

For the path $a - b - c$, we have $n = 3$.

The set $\{b\}$ is a vertex cover of size 1. Its complement

$$\{a, c\}$$

is an independent set of size

$$3 - 1 = 2.$$

So the yes-instance

$$\langle G, 1 \rangle \in \text{VERTEX-COVER}$$

corresponds to

$$\langle G, 2 \rangle \in \text{INDEPENDENT-SET}.$$

Write This in Exam: VERTEX-COVER and INDEPENDENT-SET

For any graph $G = (V, E)$ and $C \subseteq V$,

$$C \text{ is a vertex cover} \iff V - C \text{ is an independent set.}$$

This is because an uncovered edge would have both endpoints outside C , and that is exactly an

edge inside $V - C$.

To reduce VERTEX-COVER to INDEPENDENT-SET, take an instance $\langle G, k \rangle$ with $n = |V|$ and output

$$\langle G, n - k \rangle.$$

Then

$$G \text{ has a vertex cover of size } \leq k \iff G \text{ has an independent set of size } \geq n - k.$$

To reduce INDEPENDENT-SET to VERTEX-COVER, take $\langle G, k \rangle$ and output

$$\langle G, n - k \rangle.$$

Then

$$G \text{ has an independent set of size } \geq k \iff G \text{ has a vertex cover of size } \leq n - k.$$

Both reductions are polynomial because they only change the parameter.

4 Independent Set and Clique

Complement Graph

For a graph $G = (V, E)$, the complement graph $\overline{G}$ has the same vertex set, and

$$(u, v) \in E(\overline{G}) \iff u \neq v \text{ and } (u, v) \notin E(G).$$

Key Lemma

For every $S \subseteq V(G)$,

$$S \text{ is independent in } G \iff S \text{ is a clique in } \overline{G}.$$

Proof of the Lemma

S is independent in G iff no two vertices of S are adjacent in G . By definition of $\overline{G}$, this means every two vertices of S are adjacent in $\overline{G}$. That is exactly the definition of a clique in $\overline{G}$.

4.1 INDEPENDENT-SET Reduces to CLIQUE

Reduction and Iff Proof

Given an INDEPENDENT-SET instance $\langle G, k \rangle$, construct the complement graph $\overline{G}$ and output $\langle \overline{G}, k \rangle$.

By the lemma,

G has an independent set of size at least k

iff

$\overline{G}$ has a clique of size at least k .

Therefore

$$\langle G, k \rangle \in \text{INDEPENDENT-SET} \iff \langle \overline{G}, k \rangle \in \text{CLIQUE}.$$

The complement graph is computable in polynomial time.

4.2 CLIQUE Reduces to INDEPENDENT-SET**Reduction and Iff Proof**

Given a CLIQUE instance $\langle G, k \rangle$, construct $\overline{G}$ and output

$$\langle \overline{G}, k \rangle.$$

A set is a clique in G iff it is independent in $\overline{G}$. Hence

$$\langle G, k \rangle \in \text{CLIQUE} \iff \langle \overline{G}, k \rangle \in \text{INDEPENDENT-SET}.$$

Again, the construction is polynomial time.

Example

Let G be the path $a - b - c$. The set

$$\{a, c\}$$

is independent in G . In the complement graph $\overline{G}$, the edge (a, c) exists, so $\{a, c\}$ is a clique in $\overline{G}$.

Write This in Exam: INDEPENDENT-SET and CLIQUE

Let $\overline{G}$ be the complement graph of G . For any set $S \subseteq V$,

$$S \text{ is independent in } G \iff S \text{ is a clique in } \overline{G}.$$

This is because nonedges in G become edges in $\overline{G}$.

To reduce INDEPENDENT-SET to CLIQUE, map

$$\langle G, k \rangle \mapsto \langle \overline{G}, k \rangle.$$

Then

$$\langle G, k \rangle \in \text{INDEPENDENT-SET} \iff \langle \overline{G}, k \rangle \in \text{CLIQUE}.$$

To reduce CLIQUE to INDEPENDENT-SET, use the same map:

$$\langle G, k \rangle \mapsto \langle \overline{G}, k \rangle.$$

Then

$$\langle G, k \rangle \in \text{CLIQUE} \iff \langle \overline{G}, k \rangle \in \text{INDEPENDENT-SET}.$$

Constructing $\overline{G}$ takes polynomial time.

5 Set Cover and Vertex Cover

Important Direction

The standard direct reduction is

$$\text{VERTEX-COVER} \leq_p \text{SET-COVER}.$$

This proves that SET-COVER is at least as hard as VERTEX-COVER. Since VERTEX-COVER is **NP**-complete and SET-COVER is in **NP**, this proves SET-COVER is **NP**-complete. After that, SET-COVER and VERTEX-COVER are polynomially equivalent by NP-completeness and transitivity.

Reduction Lemma

Every VERTEX-COVER instance

$$\langle G = (V, E), k \rangle$$

can be converted in polynomial time into a SET-COVER instance

$$\langle U, \mathcal{S}, k \rangle$$

such that

$$\langle G, k \rangle \in \text{VERTEX-COVER} \iff \langle U, \mathcal{S}, k \rangle \in \text{SET-COVER}.$$

Construction

Given $G = (V, E)$, define the universe to be the edge set:

$$U = E.$$

For each vertex $v \in V$, create a set

$$S_v = \{e \in E \mid e \text{ is incident on } v\}.$$

Let

$$\mathcal{S} = \{S_v \mid v \in V\}.$$

Keep the same parameter k .

Iff Correctness

First suppose $\langle G, k \rangle \in \text{VERTEX-COVER}$. Then G has a vertex cover C with $|C| \leq k$. Choose the sets

$$\{S_v \mid v \in C\}.$$

Every edge $e \in E$ is incident on some vertex $v \in C$, because C is a vertex cover. Hence $e \in S_v$, so the chosen sets cover all elements of $U = E$. Therefore

$$\langle U, \mathcal{S}, k \rangle \in \text{SET-COVER}.$$

Conversely, suppose $\langle U, \mathcal{S}, k \rangle \in \text{SET-COVER}$. Then there are at most k sets

$$S_{v_1}, S_{v_2}, \dots, S_{v_r} \quad \text{with} \quad r \leq k$$

that cover $U = E$. Let

$$C = \{v_1, v_2, \dots, v_r\}.$$

For every edge $e \in E$, since e is covered, it belongs to some set S_{v_i} . By definition of S_{v_i} , edge e is incident on v_i . Thus every edge has an endpoint in C , so C is a vertex cover of size at most k . Therefore

$$\langle G, k \rangle \in \text{VERTEX-COVER}.$$

Example

Let G have

$$V = \{a, b, c\}, \quad E = \{ab, bc\}.$$

For SET-COVER, set

$$U = \{ab, bc\}.$$

The vertex sets are

$$S_a = \{ab\}, \quad S_b = \{ab, bc\}, \quad S_c = \{bc\}.$$

Choosing vertex b as a vertex cover corresponds to choosing set S_b , which covers all elements of U .

Write This in Exam: VERTEX-COVER to SET-COVER

Given a VERTEX-COVER instance $\langle G = (V, E), k \rangle$, make a SET-COVER instance as follows:

$$U = E, \quad S_v = \{e \in E \mid e \text{ is incident on } v\} \text{ for each } v \in V,$$

and keep the same k .

If C is a vertex cover of size at most k , then the sets

$$\{S_v \mid v \in C\}$$

cover every edge in U , so they form a set cover of size at most k .

Conversely, if sets $S_{v_1}, \dots, S_{v_r}$ with $r \leq k$ cover $U = E$, then

$$C = \{v_1, \dots, v_r\}$$

touches every edge of G , so C is a vertex cover of size at most k . Therefore

$$\langle G, k \rangle \in \text{VERTEX-COVER} \iff \langle U, \mathcal{S}, k \rangle \in \text{SET-COVER}.$$

The construction is polynomial.

About “Equally Hard”

The direct standard proof is $\text{VERTEX-COVER} \leq_p \text{SET-COVER}$. This is the proof normally written in exams to show SET-COVER is NP-hard. If the question says “equally hard,” then after showing both problems are **NP**-complete, polynomial equivalence follows by the general fact that all NP-complete problems reduce to each other.

6 3-SAT and Independent Set

Reduction Lemma

Every 3-SAT instance ϕ with m clauses can be converted in polynomial time into an INDEPENDENT-SET instance

$$\langle G, m \rangle$$

such that

$$\phi \in \text{3-SAT} \iff \langle G, m \rangle \in \text{INDEPENDENT-SET}.$$

Construction

Let

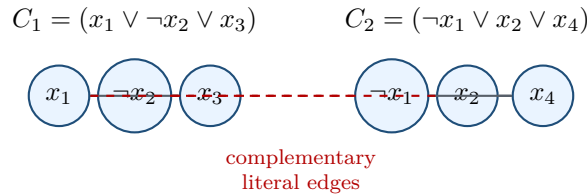
$$\phi = C_1 \wedge C_2 \wedge \cdots \wedge C_m$$

be a 3-CNF formula.

Build a graph G as follows.

1. For each literal occurrence in each clause, create one vertex. Thus each clause gives three vertices.
2. For the three vertices belonging to the same clause, add all three edges between them. So each clause becomes a triangle.
3. If two vertices from different clauses represent complementary literals, such as x_i and $\neg x_i$, add an edge between them.
4. Set

$$k = m.$$



Iff Correctness

We prove

$$\phi \in \text{3-SAT} \iff \langle G, m \rangle \in \text{INDEPENDENT-SET}.$$

First suppose ϕ is satisfiable. In each clause C_j , choose one literal that is true under the satisfying assignment. This gives one chosen vertex from each clause, so m vertices total.

These chosen vertices are independent:

- no two are from the same clause, so no triangle edge is chosen;
- no two chosen literals contradict each other, since they are all true under one assignment.

Therefore G has an independent set of size m .

Conversely, suppose G has an independent set S of size m . Since each clause triangle allows at most one chosen vertex, and there are m clauses, S must choose exactly one vertex from each clause. Also, S cannot contain two complementary literals, because such vertices are connected by an edge.

Assign all selected literals to be true. This is consistent, and every clause has one selected true literal. Hence ϕ is satisfiable.

Example

For

$$\phi = (x_1 \vee \neg x_2 \vee x_3) \wedge (\neg x_1 \vee x_2 \vee x_4),$$

we have $m = 2$. The graph has two triangles. A satisfying assignment such as

$$x_1 = 1, \quad x_2 = 1$$

allows us to choose the vertex x_1 from the first triangle and the vertex x_2 from the second triangle. These two chosen vertices are not adjacent, so they form an independent set of size 2.

Write This in Exam: 3-SAT to INDEPENDENT-SET

Let

$$\phi = C_1 \wedge C_2 \wedge \cdots \wedge C_m$$

be a 3-CNF formula. Construct a graph G as follows:

- make one vertex for each literal occurrence;
- connect the three vertices in each clause into a triangle;
- connect complementary literals from different clauses;
- set $k = m$.

If ϕ is satisfiable, choose one true literal from each clause. These m vertices are independent because they are from different clause triangles and no two chosen true literals contradict each other.

Conversely, if G has an independent set of size m , it must choose exactly one vertex from each clause triangle. Since complementary literals are joined by edges, the chosen literals are consistent. Set them true. Then every clause has a true literal, so ϕ is satisfiable.

Thus

$$\phi \in \text{3-SAT} \iff \langle G, m \rangle \in \text{INDEPENDENT-SET}.$$

The construction is polynomial because it creates $3m$ vertices and only polynomially many edges.

About the Reverse Direction

For proving NP-hardness of INDEPENDENT-SET, the needed direction is

$$\text{3-SAT} \leq_p \text{INDEPENDENT-SET}.$$

The reverse direction $\text{INDEPENDENT-SET} \leq_p \text{3-SAT}$ follows from the general theorem that 3-SAT is NP-complete and INDEPENDENT-SET is in **NP**, but it is not usually the direct construction expected in this proof.

7 Final Exam Summary

Most Important Lemmas

Pair		Key idea
VERTEX-COVER INDEPENDENT-SET	and	C is a vertex cover iff $V - C$ is independent. Parameter changes from k to $n - k$.
INDEPENDENT-SET CLIQUE	and	S is independent in G iff S is a clique in $\overline{G}$. Parameter stays k .
VERTEX-COVER SET-COVER	and	Use universe $U = E$ and one set S_v per vertex containing the edges incident on v . Parameter stays k .
3-SAT INDEPENDENT-SET	and	Make one triangle per clause and add edges between complementary literals. Target is $k = m$.

What to Write in an Exam

For each reduction, write:

1. define the source instance and target instance;
2. state the parameter k in the target instance;
3. prove the forward direction;
4. prove the backward direction;
5. say the construction is polynomial time.